

FINAL EXAM SOLUTION
DS2001-IDS

Question 1: MCSQs

1. b	11. b
2. a	12. d
3. a	13. d
4. a	14. c
5. c	15. c
6. b	16. c
7. c	17. c
8. a	18. c
9. c	19. D
10. c	20. d

Question 2A :

Cluster	Avg Spending	Transactions	Days Between	Account Age
C1	1.000	1.000	0.000	1.000
C2	0.000	0.000	1.000	0.000
C3	0.148	0.485	0.227	0.357

	C1	C2	C3
C1	0	2.000	1.207
C2	2.000	0	0.990
C3	1.207	0.990	0

First Pair to Merge->Smallest distance: 0.990 → merge C2 and C3

Compute distance to C1 using different linkages:

Single linkage: min distance between clusters

$\min(d(C1-C2), d(C1-C3)) = \min(2.000, 1.207) = 1.207$

Complete linkage: max distance between clusters

$\max(d(C1-C2), d(C1-C3)) = \max(2.000, 1.207) = 2.000$

Average linkage: average distance between all points

$$\frac{d(C1 - C2) + d(C1 - C3)}{2} = \frac{2.000 + 1.207}{2} = 1.604$$

Linkage	Characteristics	Suitability for Customer Segmentation
Single Linkage	Merges based on closest pair; can produce "chains"	May produce long, unbalanced clusters; sensitive to noise
Complete Linkage	Merges based on farthest pair; produces compact clusters	Avoids chaining; clusters are tighter and well-separated
Average Linkage	Merges based on average distance; compromise	Balanced approach; less sensitive to outliers than single linkage

Question 2B :

Principal Component Analysis (PCA) Objective: PCA is an unsupervised dimensionality reduction technique. It aims to project the data onto a lower-dimensional space such that the variance of the projected data is maximized. Essentially, it finds the directions (principal components) that capture the most information (variance) in the dataset. Does not use class labels. Helps in data compression, noise reduction, and visualization.

Linear Discriminant Analysis (LDA) Objective: LDA is a supervised dimensionality reduction and classification technique. It aims to find a linear combination of features that maximizes the separation (discriminability) between classes while minimizing the variance within each class. Uses class labels to compute between-class and within-class scatter. Often used for feature reduction before classification.

Question 3A:

Step 1: Choose a split -> Since Age is numerical, we usually consider splits between consecutive sorted values: 20, 30, 40, 50.

Possible splits:

Split 1: Age ≤ 20 / Age > 20

Split 2: Age ≤ 30 / Age > 30

Split 3: Age ≤ 40 / Age > 40

Split 4: Age ≤ 50 / Age > 50

Split: Age ≤ 30 / Age > 30

Left node (Age ≤ 30): rows with Age 20, 30 $\rightarrow$ Happy: No, Yes

Count: 2

Proportion Yes = $1/2 = 0.5$, No = $1/2 = 0.5$

Gini(left) = $1 - (0.5^2 + 0.5^2) = 1 - (0.25 + 0.25) = 0.5$

Right node (Age > 30): rows with Age 40, 50, 50 $\rightarrow$ Happy: No, Yes, Yes

Count: 3

Proportion Yes = $2/3 \approx 0.667$, No = $1/3 \approx 0.333$

Gini(right) = $1 - (0.667^2 + 0.333^2) = 1 - (0.444 + 0.111) \approx 0.445$

Weighted Gini: GINI for Age ≤ 30 split ≈ 0.467

Similarly, you can compute for other splits; typically the **best split** is chosen as the one **minimizing Gini**.

Question 3B:

$$P(\text{Happy} = \text{Yes}) = \frac{\# \text{Yes}}{\text{Total}} = \frac{3}{5} = 0.6$$

$$P(\text{Happy} = \text{No}) = \frac{\# \text{No}}{5} = \frac{2}{5} = 0.4$$

We have 3 Age categories (Younger, Adult, Elder) and 2 Occupations (CTO, Prof). Use Laplace smoothing to avoid zero probabilities.

$$P(\text{Age} = \text{Younger} | \text{Yes}) = \frac{0+1}{3+3} = \frac{1}{6} \approx 0.167$$

$$P(\text{Age} = \text{Adult} | \text{Yes}) = \frac{1+1}{3+3} = \frac{2}{6} = 0.333$$

$$P(\text{Age} = \text{Elder} | \text{Yes}) = \frac{2+1}{3+3} = \frac{3}{6} = 0.5$$

$$P(\text{Age} = \text{Younger} | \text{No}) = \frac{1+1}{2+3} = \frac{2}{5} = 0.4$$

$$P(\text{Age} = \text{Adult} | \text{No}) = \frac{0+1}{2+3} = \frac{1}{5} = 0.2$$

$$P(\text{Age} = \text{Elder} | \text{No}) = \frac{1+1}{2+3} = \frac{2}{5} = 0.4$$

$$P(\text{CTO} | \text{Yes}) = \frac{1+1}{3+2} = \frac{2}{5} = 0.4$$

$$P(\text{Prof} | \text{Yes}) = \frac{2+1}{3+2} = \frac{3}{5} = 0.6$$

$$P(\text{CTO} | \text{No}) = \frac{1+1}{2+2} = \frac{2}{4} = 0.5$$

$$P(\text{Prof} | \text{No}) = \frac{1+1}{2+2} = \frac{2}{4} = 0.5$$

$$P(\text{Yes} | X) \propto P(\text{Yes}) \cdot P(\text{Age} = \text{Younger} | \text{Yes}) \cdot P(\text{Occupation} = \text{Prof} | \text{Yes})$$

$$P(\text{Yes} | X) = 0.6 * 0.167 * 0.6 \approx 0.060$$

$$P(\text{No} | X) \propto P(\text{No}) \cdot P(\text{Age} = \text{Younger} | \text{No}) \cdot P(\text{Occupation} = \text{Prof} | \text{No})$$

$$P(\text{No} | X) = 0.4 * 0.4 * 0.5 = 0.08$$

$P(\text{No} | X) = 0.08 > P(\text{Yes} | X) = 0.060$ predicted class: No

Question 4A:

Apriori Principle:

If an itemset is frequent, then all its subsets must also be frequent.

Conversely, if an itemset is infrequent, then all its supersets are also infrequent.

This principle helps reduce the number of candidate itemsets in frequent itemset mining.

For example, if {Bread, Milk} is infrequent, there's no need to check {Bread, Milk, Eggs}. This drastically prunes the search space, making the algorithm efficient.

Question 4B:

Frequent 1-itemsets: {Bread, Milk, Butter}

Frequent 2-itemsets: {Bread, Milk}

Question 4C:

Support = (# transactions containing both Coffee & Sugar) / total transactions =
30/10000=0.3%

Confidence = (# transactions containing both Coffee & Sugar) / (# transactions containing Coffee) = $30/50 = 0.6 = 60\%$

Bread → Milk has highest support (35%), making it most valuable for promotions, since it applies to many customers.

Question 5A:

- I. Mean accuracy = 86.2%, SD $\approx 2.14\%$
- II. Total positive cases = 120
10-fold CV → 10 folds
Positive cases per fold = $120/10 = 12$
Each fold should contain 12 positive cases.
- III. If training the model on N samples takes time $O(f(N))$, LOOCV requires $N \times O(f(N))$, which is computationally expensive for large datasets.
For 1,200 patients: LOOCV iterations = 1,200

Question 5B:

Metric/Class	Class L (Legitimate)	Class S (Shady)	Class F (Fraud)
Precision	$720/760 = 0.947$	$90/150 = 0.6$	$40/90 = 0.444$
Recall	$720/780 = 0.923$	$90/150 = 0.6$	$40/70 = 0.571$
F1-Score	0.934	0.6	0.5
Accuracy	$850/1000 = 85\%$		

- I. above
- II. False Positive (FP): Predicting a transaction as Fraud when it is actually Legitimate. Bank might block a legitimate customer's transaction, causing customer dissatisfaction.
False Negative (FN): Predicting a transaction as Legitimate when it is actually Fraud. Bank loses money, may face financial loss, regulatory issues, or fraud liability.
More costly: False Negative, because failing to detect fraud directly impacts the bank's finances, whereas false positives mostly cause inconvenience.
- III. The dataset is highly imbalanced: 85% legitimate, 15% fraud (approx). A model can predict all transactions as legitimate and still achieve high accuracy. Example: Predict all as legitimate → accuracy = 85%. In the given scenario: 99.8% accuracy, but only 45% of frauds detected → model fails to catch many fraudulent transactions.
Conclusion: High accuracy is misleading in imbalanced datasets because it does not reflect the model's ability to detect the minority (fraud) class.

Better Evaluation Metrics:

- ✓ Precision (Fraud class): How many predicted frauds are actually fraud.
- ✓ Recall / Sensitivity (Fraud class): How many actual frauds are detected.
- ✓ F1-Score: Harmonic mean of precision and recall, balances detection and false alarms.
- ✓ ROC-AUC / PR-AUC: Evaluate model performance across thresholds, especially for imbalanced classes.